



HK IMO Training 2018-2019
Problem Set 4
Date due: 13 October 2018



Problem 17. The director has found out that six conspiracies have been set up in his department, each of them involving exactly three persons. Prove that the director can split the department into two laboratories so that none of the conspiring groups is entirely in the same laboratory.

Problem 18. A photographer took some pictures at a party with 10 people. Each of the 45 possible pairs of people appears together on exactly one photo, and each photo shows/depicts two or three people. What is the smallest possible number of photos taken?

Problem 19. Let $ABCD$ be a parallelogram. The circle with diameter AC intersects the line BD at points P and Q . The perpendicular to the line AC passing through the point C intersects the lines AB and AD at points X and Y , respectively. Prove that the points P, Q, X and Y lie on the same circle.

Problem 20. Determine all positive integers $n \geq 3$ such that the inequality

$$a_1a_2 + a_2a_3 + \dots + a_{n-1}a_n + a_na_1 \leq 0$$

holds for all real numbers $a_1, a_2, \dots, a_n$ which satisfy $a_1 + a_2 + \dots + a_n = 0$.